Topic 7—Lecture Notes

Introduction to Isoparametric Concept & Element Quality

Computational Statics & Dynamics (7252ENG)

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Isoparametric Elements
Natural and Physical (Local/Global) Coordinates
Definition of Jacobian Matrix
Isoparameteric vs Generic FE Flow
Mesh
Element Quality
Refinement Strategies
Convergence in Practice: u vs σ
References

Part I

Complexity, Error Analysis, and Verification & Validation

Isoparametric Elements

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Isoparametric Mapping: Natural, Physical, and Global Coordinates

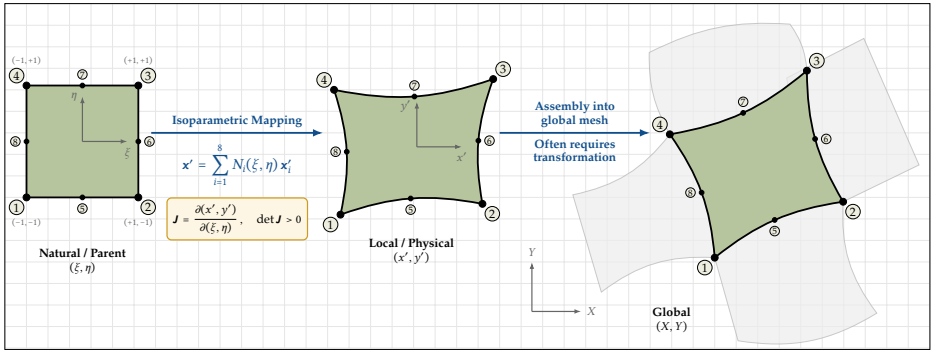


Figure 1. Natural (ξ, η), local/physical (x', y'), and global (X, Y) coordinates, connected by the isoparametric mapping and the Jacobian.

Definition 1.1 — Isoparametric Concept. The same shape functions $N_i(\xi, \eta)$ interpolate both geometry and the displacement field:

$$x = \sum_i N_i(\xi, \eta)\, x_i, \qquad u = \sum_i N_i(\xi, \eta)\, u_i$$

mapping the fixed reference (parent) element onto the arbitrary physical shape of each element, which is then placed into the assembled global mesh.

Keypoint 1.1 — Role of the Jacobian. The Jacobian $J = \partial(x, y)/\partial(\xi, \eta)$ is the link between natural and physical space: it converts derivatives ($B = J^{-1} \nabla_\xi N$) and differential area/volume ($dA = \det(J)\, d\xi\, d\eta$).

Definition of Jacobian Matrix

Definition 1.2 — Jacobian Matrix (Cook et al. 2002). For an element with shape functions $N_i(\xi, \eta)$ and nodal coordinates (x_i, y_i), the Jacobian is

$$\boldsymbol{J} = \frac{\partial(x, y)}{\partial(\xi, \eta)} = \begin{bmatrix} \dfrac{\partial x}{\partial \xi} & \dfrac{\partial y}{\partial \xi} \\ \dfrac{\partial x}{\partial \eta} & \dfrac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \displaystyle\sum_i \frac{\partial N_i}{\partial \xi} x_i & \displaystyle\sum_i \frac{\partial N_i}{\partial \xi} y_i \\ \displaystyle\sum_i \frac{\partial N_i}{\partial \eta} x_i & \displaystyle\sum_i \frac{\partial N_i}{\partial \eta} y_i \end{bmatrix}$$

i.e., a purely local quantity built from shape-function derivatives (known functions of ξ, η) and the element's own nodal coordinates.

Keypoint 1.2 — Uniformity of Jacobian. Distortions of an element, i.e., the deviations from an ideal reference element (equilateral triangle, unit square), is captured by various metrics to characterise the followings within an element:

1. **Jacobian ratio:** the uniformity of Jacobian ($\min/\max > 0.3$ prevents inaccuracies in stres/strain calculations); and
2. **Jacobian determinant:** the determinant of Jacobian ($\det(\boldsymbol{J}) > 0$ prevents collapsed or inverted elements).

Example 1.1 Non-uniformity of Jacobian matrix is caused by its dependency on the local coordinates; it happens in the following cases:

1. curved edges,
2. non-parallel sides, or
3. misplaced midside nodes.

A mild variation is ok but drastic changes cause inaccuracies in gradient recovery.

Application in iso-p formulation

- **Differentiation:** N_i is defined in (ξ, η), but strains need $\partial N_i/\partial x, \partial N_i/\partial y$:

$$\begin{Bmatrix} \partial N_i/\partial x \\ \partial N_i/\partial y \end{Bmatrix} = \boldsymbol{J}^{-1} \begin{Bmatrix} \partial N_i/\partial \xi \\ \partial N_i/\partial \eta \end{Bmatrix}$$

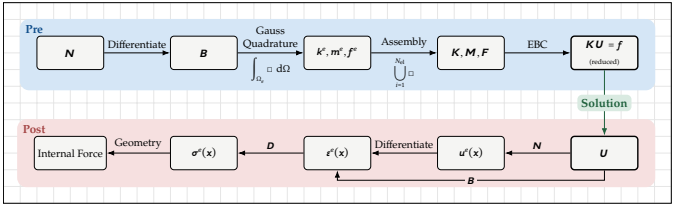
  i.e., this is how the strain-displacement matrix $\boldsymbol{B}$ is assembled. $\boldsymbol{J}^{-1}$ exists only if $\det(\boldsymbol{J}) \neq 0$; a valid, non-inverted element further requires $\det(\boldsymbol{J}) > 0$ at every point.

- **Differential area/volume:**

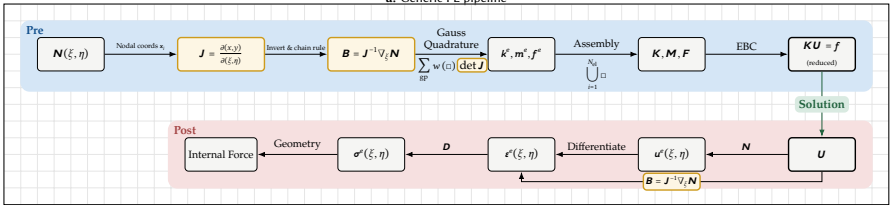
$$\mathrm{d}x\,\mathrm{d}y = \det(\boldsymbol{J})\,\mathrm{d}\xi\,\mathrm{d}\eta$$

  $\det(\boldsymbol{J})$ is the local area/volume scale factor used when integrating $\boldsymbol{k}^e, \boldsymbol{m}^e, \boldsymbol{f}^e$ over the parent domain.

Isoparameteric vs Generic FE Flow



a. Generic FE pipeline



b. Isoparametric FE pipeline

Figure 2. (a) vs. (b): $\boldsymbol{N}$ picks up an explicit (ξ, η) argument and a new $\boldsymbol{J}$ step is inserted before $\boldsymbol{B}$. Assembly and solution are identical but in recovery, Jacobian is involved again.

Keypoint 1.3 — Two Uses of the Jacobian. $\boldsymbol{J}$ appears twice: via $\boldsymbol{J}^{-1}$ to build $\boldsymbol{B}$ for the strain-displacement relation, and via $\det(\boldsymbol{J})$ to weight the Gauss-quadrature sum for $\boldsymbol{k^e}, \boldsymbol{m^e}, \boldsymbol{f^e}$, i.e., the same matrix is involved in both differentiation and integration.

Mesh

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Element Quality

Refinement Strategies

Convergence in Practice: u vs σ

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Element Quality

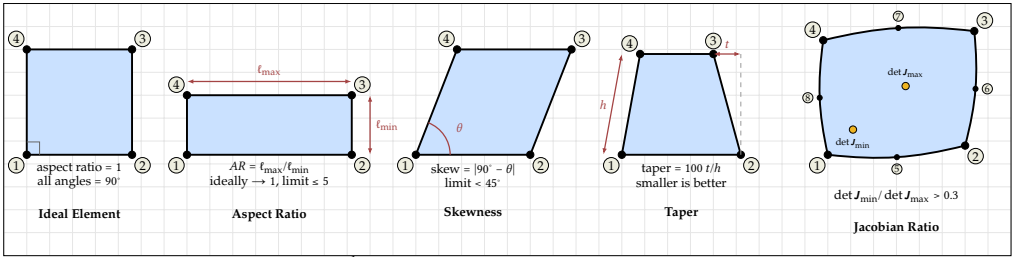


Figure 3. Five metrics for assessing element quality.

Distortion Types (Hellen & Becker 2017)

- **Stretch (bad aspect ratio):** unequal edge lengths ⇒ $\boldsymbol{J}$ badly scaled between ξ and η
- **Skew:** non-orthogonal edges ⇒ large off-diagonal terms in $\boldsymbol{J}$
- **Taper:** non-parallel edges ⇒ $\det(\boldsymbol{J})$ varies strongly over the element ⇒ Gauss-quadrature integral
- **Curvature/warping:** curved or non-planar edges → $\det(\boldsymbol{J}) \to 0$ or changes sign locally
- **Invalid mapping:** $\det(\boldsymbol{J}) \leq 0$ anywhere ⇒ inverted element, solver aborts

Non-uniformity consequences

- **Alteration of Polynomial Degrees:** a non-uniform Jacobian's inverse ($\mathbf{J}^{-1}$) turns clean polynomial displacement derivatives into complex rational functions when mapped to global strains.
- **Numerical Integration Errors:** Gauss quadrature often integrates standard polynomials exactly but cannot exactly integrate the rational functions from a varying Jacobian, injecting numerical error into the stiffness matrix.
- **Artificial Straining and Shear Locking:** highly non-uniform Jacobians struggle to represent simple states (e.g., uniform tension and pure bending), introducing spurious energy modes and artificial 'shear locking'.
- **Unstable Nodal Extrapolation:** extrapolating stresses/strains from Gauss points to nodes becomes unstable under a rapidly changing Jacobian, causing non-physical stress discontinuities between elements.

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Mesh Refinement Strategies

Definition 2.1 — h-, p-, and hp-Refinement. Classical strategies for improving FE accuracy are (Cook et al. 2002):

- **h-refinement:** reduce element size h, fixed p
- **p-refinement:** increase polynomial order p, fixed mesh
- **hp-refinement:** combine both for optimal convergence
- **r-refinesment:** mesh and order is the same, discretisation arrangement changes

Refining the mesh ($h\downarrow$) and/or raising element order ($p\uparrow$) both reduce error — smooth solutions benefit most from p; singular solutions are limited by regularity m regardless of p.

Keypoint 2.1 — Convergence Rates. For smooth solutions, p-refinement gives exponential convergence; h-refinement gives algebraic convergence, $O(h^{p+1})$ in displacement and $O(h^{p})$ in stress.

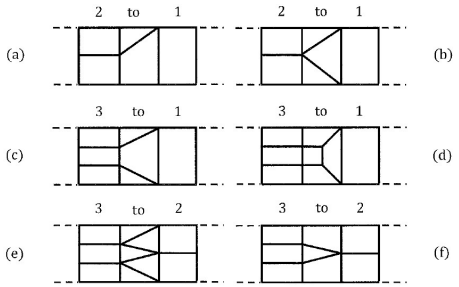
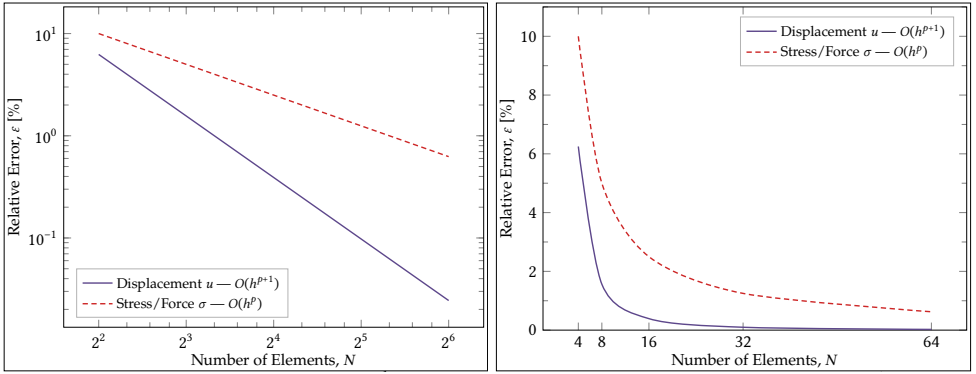


Figure 4. Graded mesh (non-uniform mesh): refinements from a coarse to fine mesh region are concentrated near a stress concentration (Hellen & Becker 2017).

Convergence in Practice: u vs σ



a. Log-log — power-law rates are straight lines

b. Linear — same data, rates visually hidden

Figure 5. Convergence of relative error with mesh refinement: displacement $O(h^{p+1})$, one order faster than stress/force $O(h^p)$ (Hughes 2000).

Keypoint 2.2 — Why Stress Lags Displacement. Stress/force is a derivative of the displacement field ($\sigma = \boldsymbol{DBu}^e$) and thus differentiating drops the interpolation order by one, so its error converges one power of h slower.

Why Plot on Log-Log Axes?

On linear axes (b) both curves look flat and "converged" after a few refinements. On log-log axes (a) the same data is a straight line of slope p or $p+1$ — the reliable way to read off convergence order.

References I

Cook, RD, Malkus, DS, Plesha, ME & Witt, RJ (2002), *Concepts and Applications of Finite Element Analysis*, 4th, John Wiley & Sons, New York.

Hellen, T & Becker, A (2017), "Mesh Generation and Element Quality," *Finite Element Analysis for Engineers: A Primer*, NAFEMS, Glasgow, UK, chap. 12.

Hughes, TJR (2000), *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*, Dover Publications, Mineola, NY.